A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
03.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** They are driven by renewable, low energy-density resources, especially solar energy, which will not be depleted: the hydrological cycle, the carbon and nitrogen cycles, atmospheric and thermohaline circulation **1** Most natural processes occur at low temperatures, because enzymes lower the activation energy of the reactions, so little energy is needed **1** Processes link into cycles in which the waste product of one is the raw material for the next, so raw materials are not depleted and wastes do not build up **1** A small number of abundant, simple elements are built into monomers and then into a wide range of polymers, so supplies of the raw materials are secure **1** Wastes are biodegradable and break down to non-toxic products, so they do not accumulate or poison the system **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **thermohaline** → the ocean conveyor driven by temperature and salinity • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals are dispersed after use - spread thinly through products all over the world, or lost to corrosion - so there is no concentrated deposit left to recover **1** Products are made of mixtures of materials that are difficult to separate, so recovery and re-use are expensive and often incomplete **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
03.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Design for extended use:** a product with a long life, which can be repaired, gives fewer opportunities for its materials to be lost **1 Design for end-of-life re-use:** materials that can be identified and separated easily are far more likely to be recycled **1** Separation of technical and biological materials, because metals and plastics need different reprocessing from biodegradable materials, which can be composted or digested **1** Energy from renewable sources at every stage, as in natural systems **1** Optimum rather than maximum production, so that the system supports the processes it relies on instead of maximising one output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sustainability depends on the total impact, which is the number of people multiplied by the impact of each person, so the same activity crosses the threshold as either rises **1** A natural process can absorb or replace what is taken only up to a rate – beyond that the resource is depleted or the waste accumulates faster than it can be broken down **1 Example:** slash and burn agriculture is sustainable where the population is small enough for each clearing to be left fallow long enough to recover, but not when land has to be re-cleared before the soil fertility has returned **1 Example:** discharging untreated sewage into a river is harmless where decomposers can break it down, but causes deoxygenation and eutrophication once the load exceeds that rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **threshold** → the concentration below which no harm is detected • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
03.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Urban populations are supplied by resources brought in from elsewhere, so the damage caused by extraction happens out of sight and often in another country **1** In a sedentary rural community the resource is local, so the problems of over-exploitation become obvious quickly as supplies dwindle, and a cultural belief in limiting exploitation to needs develops **1 Consequence:** when one area is exhausted, exploitation simply moves to another area and the consumer sees no change, so the total damage keeps growing - this has happened with timber, energy, metal minerals and fish **1 Consequence:** there is no pressure from consumers to change, so unsustainable practices continue until the resource fails on a large scale **1 Consequence:** the costs fall on the people and ecosystems where extraction happens, not on those who benefit, so the market price never reflects the true environmental cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-377; AQA spec 3.6.2-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Readings drift so they may not be accurate – calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions **1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly **1** If a meter cannot be adjusted, record the percentage error and correct the results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Same height above the ground **1** No obstacles to air flow – impeller type facing into the wind **1** Axis of rotation horizontal (impeller) or vertical (cup) – same time period / averaging **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling **1** Evaporation cools the wet bulb more when humidity is low – the temperature difference is compared with a table to find humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Test strips with reagent bands compared with a colour chart (semi-quantitative) **1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength **1** Ion-selective electrode measuring electrode potential (other ions can interfere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Marks shared** Up to 5 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds – oak supports many invertebrates **up to 2** B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies – fewer niches **up to 2** C: non-indigenous single-species, single-age conifers support few native invertebrates – dense shade – no clearings or dead wood – needles acidify soil **up to 2** Habitat diversity / structural variety increases the range of niches and hence species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	5
05.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Designated protected areas:** SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks **1 Legal protection of ancient woodland:** planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) **1 Conservation management:** coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer **1** Planting new native woodlands linking fragments (corridors) – Woodland Trust / National Trust ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** No species is completely dominant and food webs have many alternative links **1** So a change in the population of one species has a relatively small impact on the whole community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1 SSSI:** UK legislation (Wildlife and Countryside Act) – best examples of UK flora, fauna, geological features – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) **1 SPA:** EU Birds Directive – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) **1 Ramsar:** Ramsar Convention (intergovernmental agreement) – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally **1 MCZ:** UK marine legislation (Marine and Coastal Access Act) under the W&CA framework – protects marine habitats / species by restricting activities such as fishing and dredging – NNR: best SSSIs with complete communities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
06.2	**1 mark each** 1 mark for each of the 3 points below. **1 Legal status:** restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) **1** Management agreements between owners and statutory bodies to maintain suitable conditions **1** But it does not protect against external influences such as air pollution or climate change – very few sites are state-owned so protection depends on owner cooperation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues **1** Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) **1** Benefits everyone so society pays through central funding – legislation cannot compel positive management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
08.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** A larger area supports larger populations, which are less likely to be lost to chance events or inbreeding, and it can hold species with large territories such as top predators **1** A larger area holds a wider range of habitats and microhabitats, so more species can find the conditions they need **1** A compact, rounded shape has the smallest perimeter for its area, so less of the reserve is affected by edge effects such as wind, drier conditions, spray drift and disturbance from outside **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** It allows individuals to move between the two areas, so the populations interbreed as one larger gene pool and inbreeding is reduced **1** An area that loses a species can be recolonised from the other, and species can shift their range as the climate changes rather than being trapped in an isolated fragment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water levels:** sluices and bunds hold water back to create ponds, scrapes and seasonally flooded marsh, adding wetland habitats and a range of depths for different species **1 Soil nutrient levels:** cutting and removing hay rather than leaving it, and never fertilising, lowers the nutrient level so that vigorous grasses cannot out-compete the wildflowers of species-rich grassland **1 Topography:** banks, scrapes and south-facing slopes create warm and cool, wet and dry microhabitats within a small area, each suiting different species **1 Light levels:** thinning, coppicing and cutting rides and glades lets light reach the woodland floor, so ground flora and the insects that depend on it can grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** A plagioclimax is a community held at an earlier stage of succession than its natural climax by continued human activity, so it never develops into woodland **1** Grazing animals eat the tree and shrub seedlings that would otherwise shade out the grassland, so succession is deflected and the open habitat survives **1** The species, stocking density and timing of grazing are chosen to suit the habitat – cattle graze less selectively and their trampling creates bare patches for seedlings, while sheep graze close and produce a short, even sward **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3

Do not write outside the box

Question	Marking guidance	Marks
09.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invasive non-native species such as rhododendron or Himalayan balsam out-compete and shade out native species, and have no natural controls here **1** Introduced predators or competitors, such as grey squirrels or mink, reduce native populations directly **1** Without their natural predators, herbivore numbers such as deer can exceed the carrying capacity, preventing tree regeneration and damaging the ground flora **1** Culling may also be needed to keep a population within the reserve's carrying capacity so that it does not damage the habitat the whole community depends on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-49; AQA spec 3.1.3.3-3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

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